

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

EEE 315 — Past Questions & Materials
by Renance

Course: EEE 315
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200, 300

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Paper 1 of 3 — EEE 315 - 2024/2025 - First Semester - Electrical Machines

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Examination: First Semester
Session: 2024/2025
Degree: B. Eng
Date: 27/05/2025



Course Code: EEE315
Course Title: Electrical Machines I
Time Allowed: 2 Hours

INSTRUCTION: (i) Answer TWO (2) questions from each section.
(ii) Begin each question on a fresh page.

SECTION A

QUESTION ONE

- How would you briefly explain a composite series magnetic circuit? 6marks
- How would you describe briefly, the constructional features of the stationary part of a d.c machine? 4marks
- A 10kW shunt generator having an armature circuit resistance of 0.75Ω and a field resistance of 125Ω , generates a terminal voltage of 250V at full load. Determine the efficiency of the generator at full load, assuming the iron, friction and windage losses amount to 600W. 7½marks

QUESTION TWO

- With the aid of a circuit arrangement, briefly explain the separately excited generator 5marks
- In a long-shunt compound generator, the terminal voltage is 230V when generator delivers 150A. Determine:
 - Induced e.m.f.
 - total power generated and
 - distribution of this power.

Given that shunt field, series field, divertor and armature resistances are 92Ω , 0.015Ω , 0.03Ω and 0.032Ω respectively. 10marks
- Produce the load current/ voltage characteristic of a compound generator 2½marks

QUESTION THREE

- What do you understand by the term electromechanical energy conversion principle? 2½marks
- Produce the flow diagram of an electromechanical energy conversion system as studied in this course. 5marks
- Explain the losses that occur in D.C Machines. 5marks

- (d) A 4-pole generator has a lap-wound armature with 50 slots with 16 conductors per slot. The useful flux per pole is 30mWb. Determine the speed at which the machine must be driven to generate an e.m.f. of 240V. 5marks

SECTION B

QUESTION FOUR

- (a) A 400V shunt connected d.c motor takes a total current of 5.5A on no load and 59.5A at full load. The field circuit resistance is 267 ohms and the armature circuit resistance is 0.2ohms (excluding brushes where the drop may be taken as 2V). If the armature reaction effect at full-load weakens the flux per pole by 2% change in speed from no-load to full load. 9 marks
- (b) What resistance must be placed in series with the armature in the machine if the full load speed is to be reduced by 50% with the gross torque remaining constant? Assume no change in the flux. 8.5 marks

QUESTION FIVE

A 100KVA transformer has 400 turns on the primary and 80 turns on the secondary. The primary and secondary resistances are 0.3 ohms and 0.01ohms respectively. The corresponding leakage reactance are 1.1ohms and 0.035ohms respectively. The supply voltage is 2200V. Calculate

- (ai) the equivalent impedance referred to the primary 4marks
- (aii) the voltage regulation, and 4marks
- (aiii) the secondary terminal voltage for full load having a power factor of 0.8 leading. 4marks
- (b) Explain why we conduct:
- (i) Open circuit and 2.5marks
- (ii) short-circuit tests on transformers 3marks

QUESTION SIX

A 500V, 10hp shunt motor has a full-load efficiency of 85% with the same shunt field and armature current. It is desired to reduce the speed by 30% by inserting a resistance in the armature circuit. Calculate:

- (a) the value of the inserted resistance. 8.5marks

Given that the resistance of the field and armature are 400ohms and 0.25 ohms respectively.

- (b) Explain 3 methods of speed control of dc motors 9marks

Paper 2 of 3 — EEE 315 Electrical Machines I 2025

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES. EFFURUN
DEPARTMENT OF ELECTRIC AL AND ELECTRONICS ENGINEERING

Course Code: FEEAIS :

Course Title: Electrical Machines |

Time Allowed: 2 Hours :

Examination: First Semester

Session: 2024/2025

Degree: B. Ene

Date: 27:08 2024

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(b) Produce the flow diagram of an electromechanical energy conversion system as studied in this course. Marks

(c) Explain the losses that occur in D.C Machines. Marks

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SECTION B

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QUESTION FOUR

(a) A 400V shunt connected d.c motor takes a total current of 5.54 on no load and 59.5A at full load. The field circuit resistance is 267 ohms and the armature circuit resistance is 0.2ohms (excluding brushes where the drop may be taken as 2V). If the armature reaction effect at full-load weakens the flux per pole by 2% change in speed from no-load to full load. Calculate marks

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(b) Explain SO) SP ah 3

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Paper 3 of 3 — EEE 315 Electrical Machines

Level 300 • First Semester • PastQuestion

QUESTION Six (6)

6a. A 200 KVA rated transformer has a full-load copper loss of 1.5kW and an iron loss of 1 kW. Determine the transformer efficiency at 0.85 power factor, also determine the efficiency of the transformer at half full load and 0.8 (10 Marks)

6b. A series circuit consists of a 240 V AC source, a resistor of 20 Ω , and an inductor of 0.1 H. The measured current is 1.0 A and the measured power is 150 W. Find the value of X (8 Marks)

6c. Draw the diagram of a parallel connection of a series transformer

under the following conditions

1. Load is constant

2. Load is variable

(7 Marks)